Keeping Things Stable

How Controllers Tackle Complexity in Modern Physics

EXERCISE

ÖAW AUSTRIAN ACADEMY OF SCIENCES

CERN School of Computing

TU WIEN TECHNISCHE UNIVERSITÄT WIEN Vienna University of Technology

CERN

MARIETTA BLAU INSTITUTE FOR PARTICLE PHYSICS

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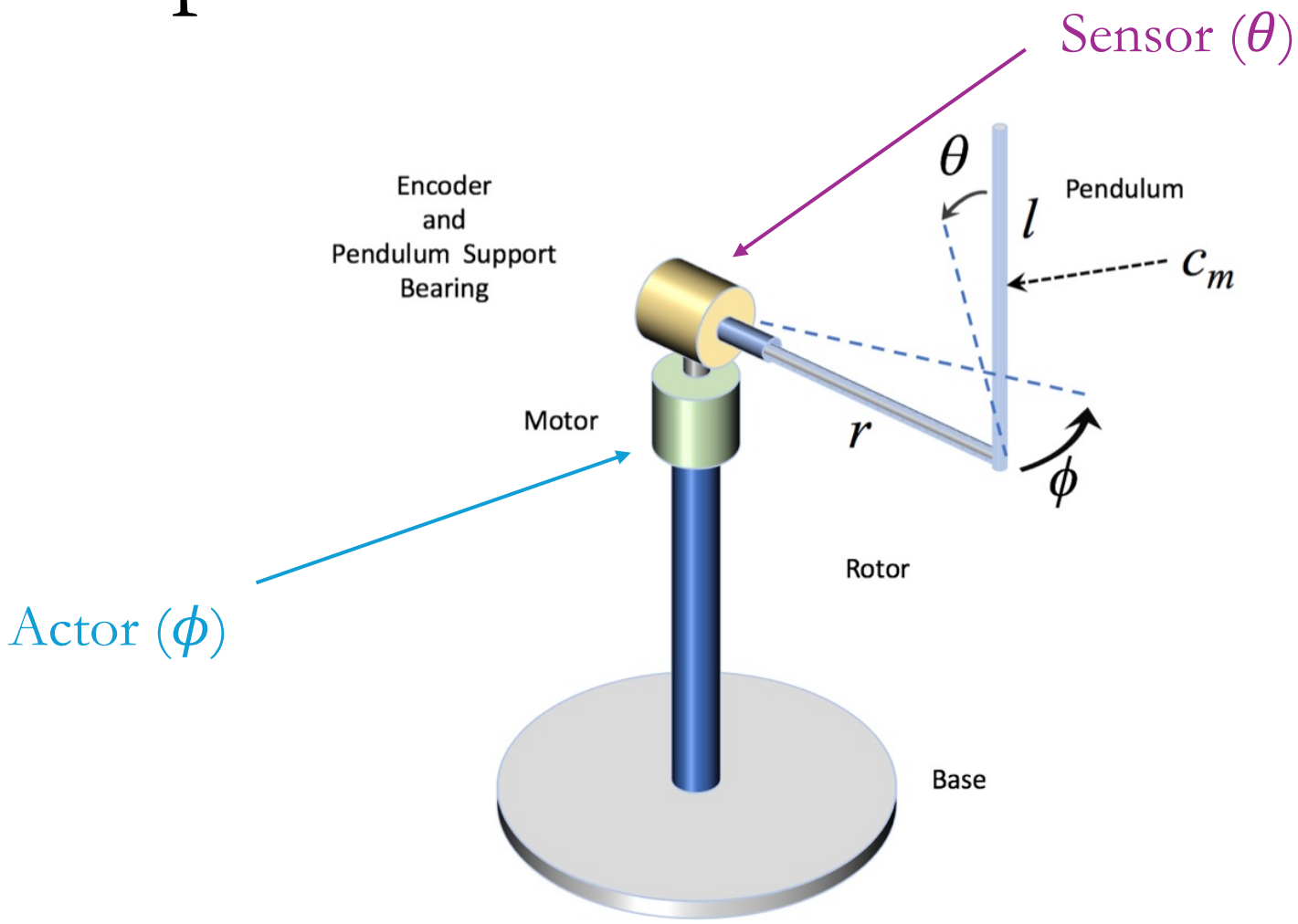
Overview

- Introduction
- Exercises
  - 1: PID
  - 2: LQR
- Competition
- Discussion

Inverted pendulum



Source: STMicroelectronics, STSW-EDUKIT01 (graphic on product page); accessed 16 Feb 2026.

Exercise procedure

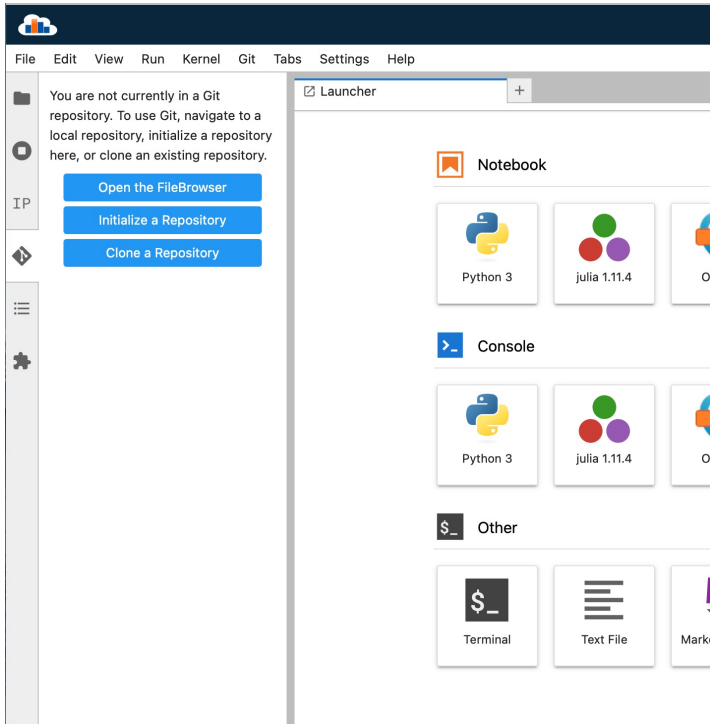
- Work in teams of 2 (max 3) people
  - Ideally: One is good in python, other one good in math.

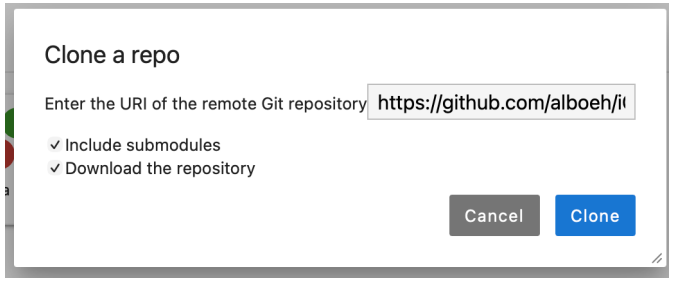
- Python notebook with a simulation of the inverted pendulum
  - It contains 2 examples to click through and 2 exercises (PID and LQR)
  - Just change code where **#TODO** is written
  - Write down your best parameters for the competition!

- GIT: https://github.com/alboeh/iCSC-2026-exercise.git

Run the notebook

- Login to SWAN: https://swan.cern.ch
  - Non-cern users can get licenses from the CSC team
- Start session
  - New JupyterLab interface
  - LCG curated stack, General purpose Software stack (109)
  - Check "Use python packages installed on CERNBox"
  - Use 4 CPU and 16 GB RAM
- Click left on Git and clone exercise rep, include submodules and download rep, **ignore error msg**
- Open exercises/InvPendulum.ipynb (**not solution**!)
  - Use "show_animation" function wisely, its time consuming
- **Alternatively**: Clone on your local machine, may be faster!





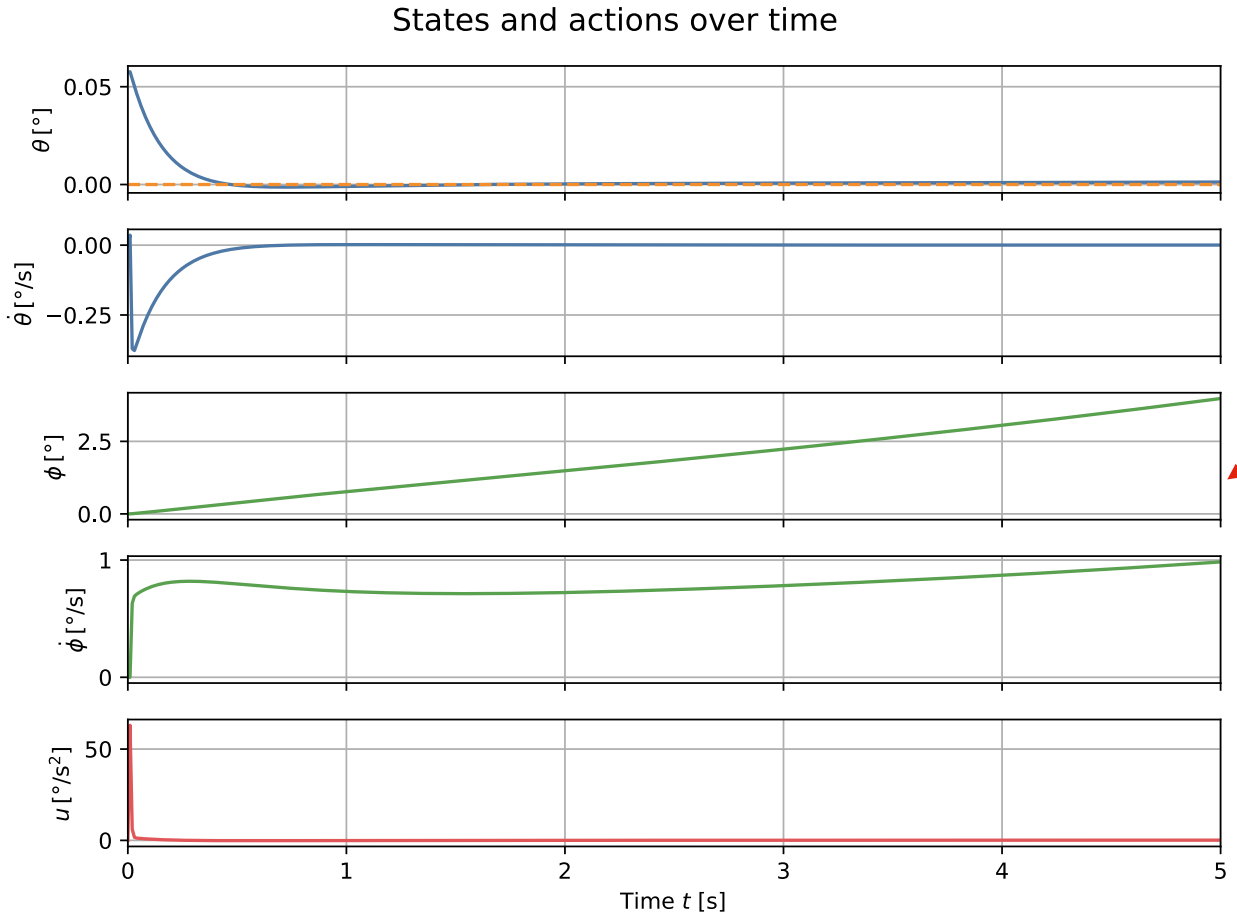
Exercise 1: PID control

$$u(t) = u_0 + \underbrace{K_c[e(t) + \frac{1}{\tau_I}\int_0^t e(t)dt + \tau_D\frac{de(t)}{dt}]}_{\Delta u}$$

Solutions Exercise 1 (PID)

There is not the "optimal" solution!

I found this PD okayish, which was $K_p = 100$ and $K_d = 10$



This is the reason we want a State-space controller!

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Exercise 2: State-Space control

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Solutions Exercise 2 (LQR)

- $$\dot{\mathbf{x}}(t) = \underbrace{\begin{pmatrix} 0 & 1 & 0 & 0 \\ \frac{3g}{2l} & -\frac{3b}{ml^2} & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}}_{\mathbf{A}} \cdot \mathbf{x}(t) + \underbrace{\begin{pmatrix} 0 \\ -\frac{3r}{2l} \\ 0 \\ 1 \end{pmatrix}}_{\mathbf{B}} \cdot u(t)$$

- Lets have a look into the code together!

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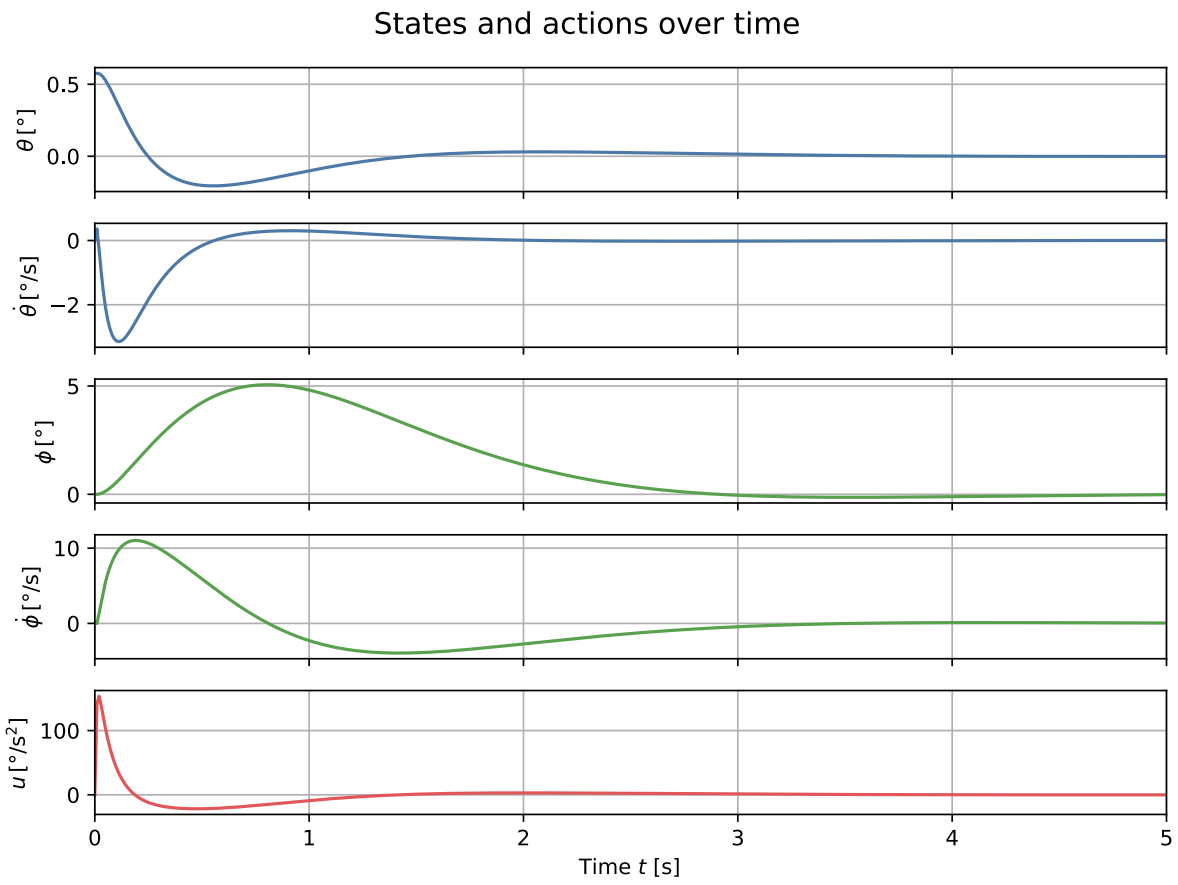
Solutions Exercise 2 (LQR)

I used
Q=(100, 1, 1000, 1)
And R=100



Everything smooth and stable now :)

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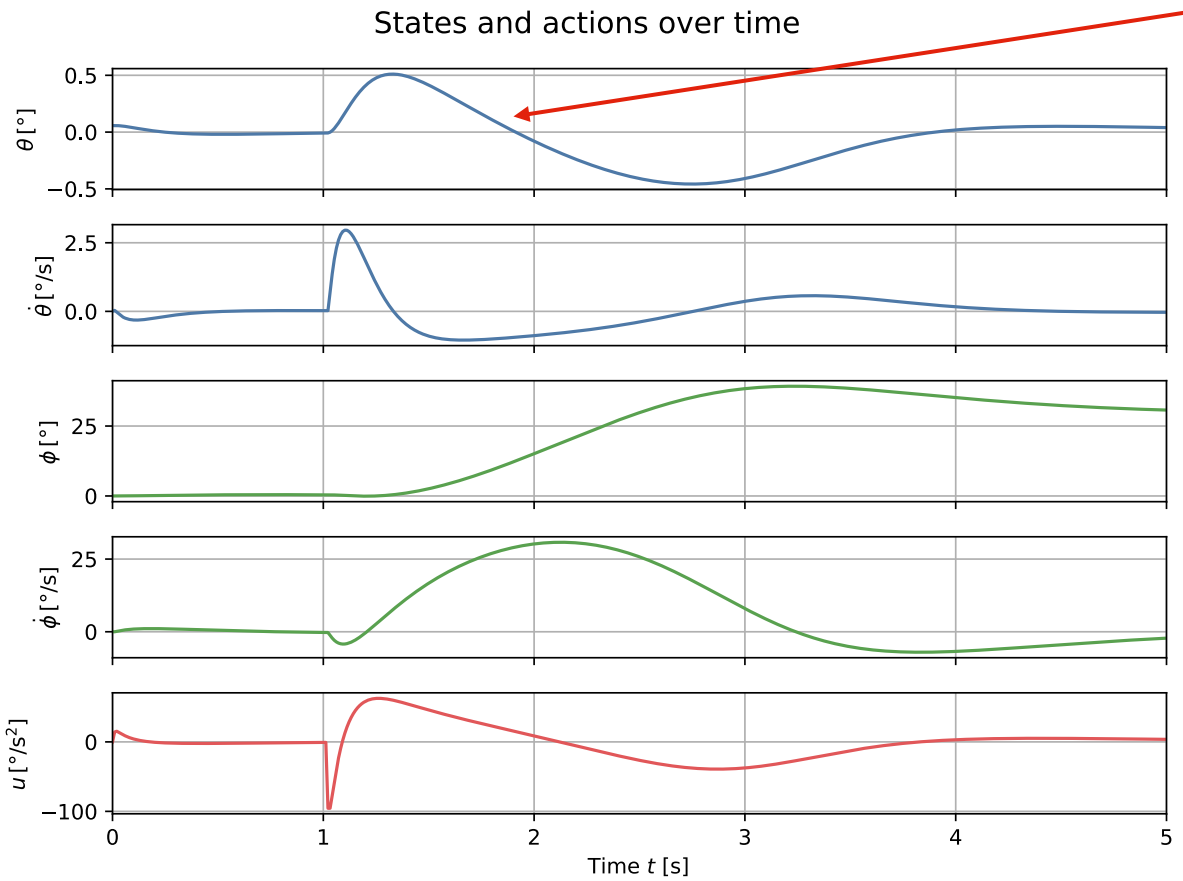
Solutions Exercise 3 (step)



We just want to change phi. Nevertheless, the controller did something „wrong“ here on purpose by counteracting the motion!
A PID would never do that, because it is „stupid“. A LQR instead knows the system and is smarter!

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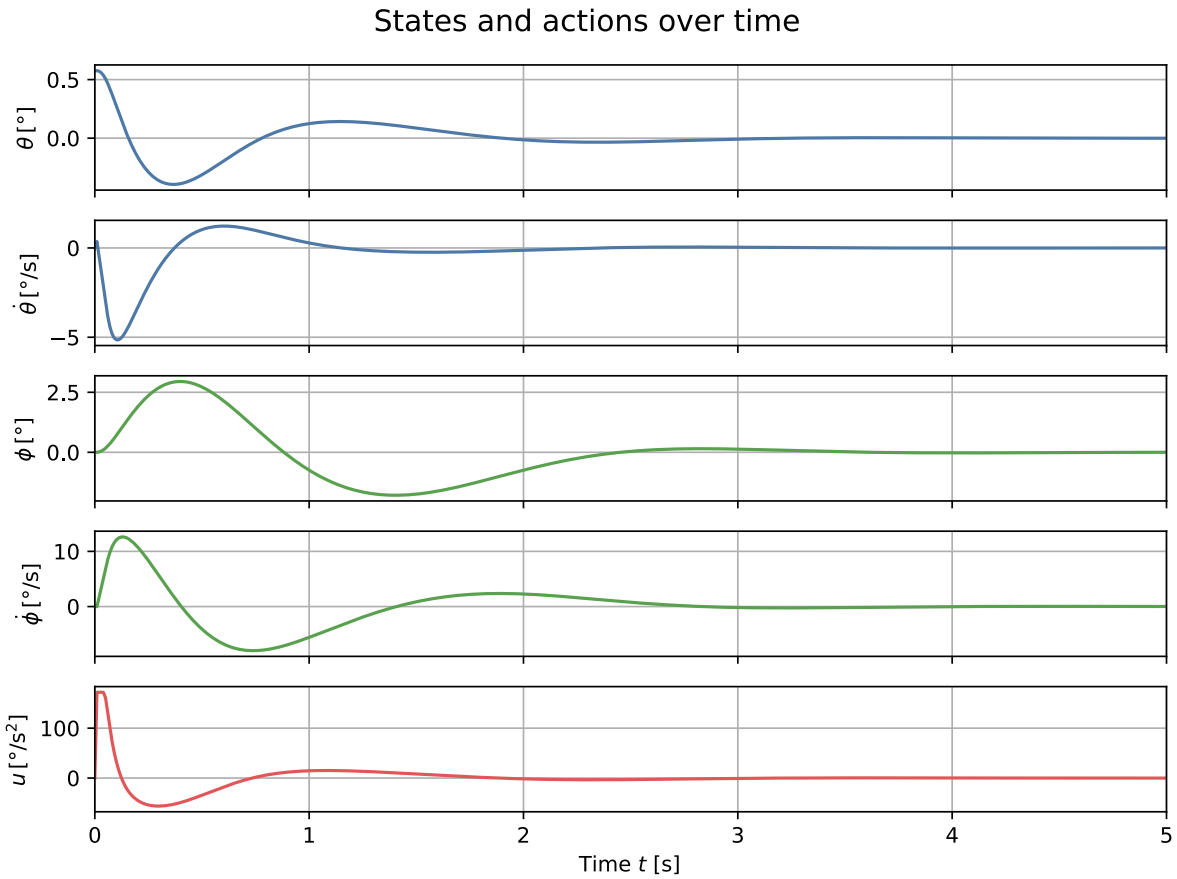
Solutions BONUS Exercise (LQI)

I used
Q=(100, 1, 1000, 1, 50000)
And R=100



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Competition

The rules

- Taking your LQR / LQI values!
- Sequence of positions (0° → 30° → -30° → 0°, each for 5s)
- Score = RMS(theta) + 1/10 * RMS(phi)
  - RMS only from 5 seconds on, so you have initially time to stabilize!
- 1x try per group during the exercise, 1 final shot!

- **Lowest score is the winner! ☺**

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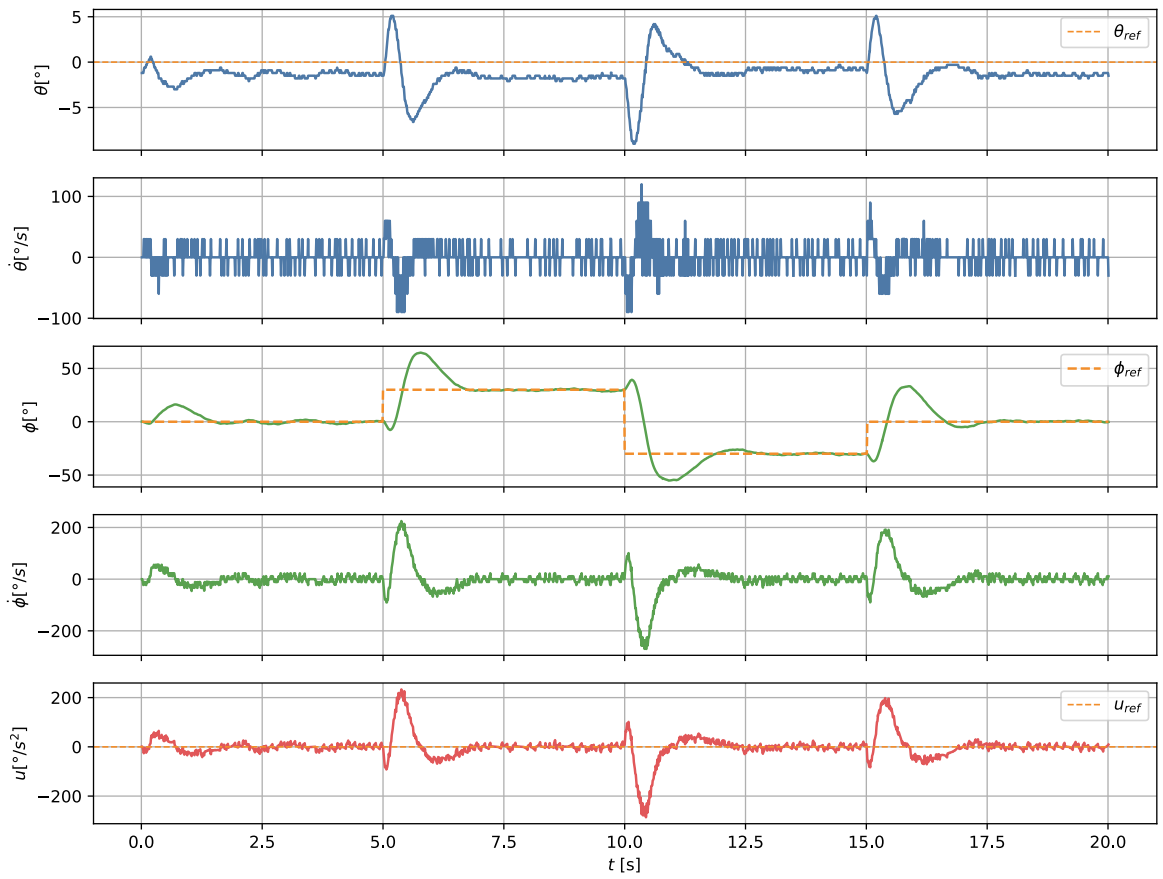
The rules

- Example

- **Lowest score is the winner! ☺**



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Discussions

Some questions…

- Is a PID or LQR better for the inverted pendulum?
- Which PID parameters and method you used? Which LQR weights?
- How much of a difference you noticed between the simulated and the real pendulum?
- Has the winning team got the “best” controller? Or were they lucky?

… and some answers

- Is a PID or LQR better for the inverted pendulum?
  - It depends in general! But here the LQR was better, it counteracted the motion
- Which PID parameters and method you used? Which LQR weights?
  - ZN is hard if system is intrinsically non-linear. PD worked best for me. I weighted theta and phi the strongest, the velocities not that much.
- How much of a difference you noticed between the simulated and the real pendulum?
  - Even though the simulation ran on non-linearized equations, it is nearly impossible to simulate it perfectly. Many things were not fully described, like the stepper dynamics, friction models, and much more!
- Has the winning team got the "best" controller? Or were they just lucky?
  - There is not something like a "best" controller. Maybe they build the "best" in keeping the pole vertically, but this is always a trade-off. E. g. maybe other groups had better results in terms of robustness (so more stable for big disturbances)

Take-aways

- PID
  - No description of system necessary, fast solution
  - Only one output can be controlled (without further ado)
  - SISSO systems

- State-Space
  - Modelling necessary
  - Therefore full state controllable

- **YES: Most of the life of a control engineer is to understand the system and find a good linearization, not the control itself ☺**

Thank you!

- https://www.youtube.com/watch?v=Rh7JuL3PRSY

- Full git project (with all plots, examples, etc)
  https://github.com/alboeh/iCSC-2026-full.git